

Pairing Moving Horizon Estimation and Model Predictive Control via Carleman Approximation for Output Feedback Control

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Abstract—An output feedback control structure is proposed for processes in the presence of disturbance and incomplete state information. It combines moving horizon estimation (MHE) and model predictive control (MPC), where Carleman approximation is employed to reduce the nonlinear process plant model and then pair and streamline the computations between the two components. After Carleman approximation, the CMHE/CMPC pair reduces the dynamic optimization problem using analytical expressions for the cost functionals and constraints. CMHE provided state estimates become the initial conditions for CMHE to decide the optimal control signals. With these signals continuously updated in the process model used in CMHE, the state estimates accuracy increases. Analytical gradient vectors and Hessian matrices are supplied to the CMHE/CMPC pair to further reduce computation expenses. We present case studies on a nonlinear CSTR system to show the improvement in computational efficiency with the proposed CMHE/CMPC pair.

I. INTRODUCTION

Model predictive control (MPC) is fast becoming the yardstick over which control system performance is evaluated. This is due to its optimality, ease of use and implicit robustness characteristics. An issue that still confounds its use is that knowledge of the process state is essential in the design of the control signals. In practice, some of the state variables (or even the controlled outputs) may not be directly measurable, or even not be physical quantities. Therefore, the need to pair estimation and control is a promising direction of research in the MPC area of control engineering. To this end, [1], [2] presented work in pairing extended Kalman filter (EKF) and nonlinear model predictive control (NMPC).

Moving horizon estimation (MHE) and MPC, as dual problems, have been widely attracting attention for their several advantageous properties, including: 1) they are readily applied to multi-input-multi-output (MIMO) systems due to their formulations; 2) they handle inequality and equality constraints in a straightforward manner due to their optimization-based nature; 3) their self-evolving policies underline the ability in dealing with plant-model mismatch. MHE uses a receding time window of outputs into the past. With limited information regarding the dynamic

model, it estimates the unmeasurable state variables and noise/disturbance values. MPC predicts the evolution of the system and computes control signals over a receding time window into the future. Expectedly, together they form a good candidate for an estimation/control pair, sharing the state-space model of the process. The state estimation obtained by MHE serves as the initial condition for the MPC to design optimal control signals. These control signals are continuously updated in the dynamic model used in MHE, completing the interconnection.

Both MHE and MPC involve solving dynamic optimization problems in real-time, which requires a considerable amount of computational effort. With increasingly larger and more complicated process models, the computations by MHE and MPC geometrically increase. If the MHE/MPC pair fails to generate an optimal solution in time, the consequent feedback delays (or suboptimal solutions) can cause performance loss or even stability issues [3]. This is one of the most significant barriers holding back the application of MHE/MPC pair.

Researchers in control engineering have proposed many approaches to reduce computational expense for MHE and MPC. The advanced-step NMHE/NMPC (as-NMHE/asNMPC) algorithms, published by Biegler and coworkers in [4][5][6][7], focuses on solving complex optimization problems off-line then performing a linear approximation in the real-time. Multi-parametric MPC developed by Pistikopoulos and coworkers in [9][10] accelerates computation via querying response hypersurfaces. Fast approaches have also been reported in applications, such as extended differential flatness approach in the MPC algorithm controlling lithium-ion battery charging [11][12][13].

Motivated by the computational issue, we propose to employ Carleman approximation to design a new MHE/MPC pair, called CMHE/CMPC. Control and optimization of bilinear systems has been extensively considered since the 70s and 80s with excellent reviews in [14][15][16]. Carleman approximation is also known as Carleman linearization [17][18][19][20][21] since the basic idea is to express a finite dimensional nonlinear system as an infinite dimensional linear one. First, a desired reference state point is chosen to perform Taylor expansion to the nonlinear system under investigation. This step naturally transforms the original state vector to a deviation from the desired reference state. Second, we expand the state vector and coefficient matrices based on the Kronecker product and introduce an augmented state vector containing the higher order deviation monomials. This results in an approximate bilinear system of larger

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dimension that carries the nonlinear dynamic information of the original system, and reduces approximation errors compared with classic linearization methods. Previous work has reported the structure of CMPC [22], further refinements in the formulation [23], and the structure and a stability analysis of CMHE [24]. This paper expands this work by proposing a CMHE/CMPC pair, combining these two algorithms. CMHE and CMPC share the same extended bilinear model, further accelerating computations. The cost functions for both CMHE and CMPC are computed analytically and analytical Gradient and Hessian matrices are supplied to facilitate the optimizations. CMHE identifies the initial condition at each receding time window for the CMPC to make optimal control decisions. These control decisions are continuously updated in the process model used by the CMHE to increase its accuracy.

II. PRELIMINARY INFORMATION

A. Nonlinear System

We consider nonlinear systems with noise in the form,

$$\dot{x} = f(x) + \sum_{j=1}^m g_j(x)u_j + \sum_{l=1}^n \gamma_l(x)w_l \quad (1)$$

$$y = h(x) + v \quad (2)$$

$$x(t_0) = x_0 \quad (3)$$

where $x \in R^{n_x}$ is the state vector, $u \in R^m$ is the vector of manipulated variables; $u = [u_1 \ u_2 \ \dots \ u_m]$, $w \in R^n$ is the vector of unknown process noise or disturbance; $w = [w_1 \ w_2 \ \dots \ w_n]$. $f(x)$ is a nonlinear vector function. $g_j(x)$, $j = 1, \dots, m$ and $\gamma_l(x)$, $l = 1, \dots, n$ are nonlinear vector functions.

Also, $y \in R^p$ is the output vector; $v \in R^p$ is the vector of unknown sensor noise. $h(x)$ is a nonlinear vector function in general; this paper focuses on a linear $h(x)$. x_0 is the initial condition at the beginning time t_0 .

B. Mathematical Preliminaries

To facilitate the introduction of the optimization formulations, we first present the Kronecker product, which is the mathematical foundation of Carleman approximation. The Kronecker product of matrix $X \in C^{N \times M}$ and matrix $Y \in C^{L \times K}$ is defined as matrix $Z \in C^{(NL) \times (MK)}$ [19].

$$X = \begin{bmatrix} x_{1,1} & x_{1,2} & \dots & x_{1,M} \\ x_{2,1} & x_{2,2} & \dots & x_{2,M} \\ \dots & \dots & \dots & \dots \\ x_{N,1} & x_{N,2} & \dots & x_{N,M} \end{bmatrix} \quad Y = \begin{bmatrix} y_{1,1} & y_{1,2} & \dots & y_{1,K} \\ y_{2,1} & y_{2,2} & \dots & y_{2,K} \\ \dots & \dots & \dots & \dots \\ y_{L,1} & y_{L,2} & \dots & y_{L,K} \end{bmatrix}$$

$$Z = X \otimes Y \equiv \begin{bmatrix} x_{1,1}y_{1,1} & \dots & x_{1,1}y_{1,K} & \dots & x_{1,M}y_{1,1} & \dots & x_{1,M}y_{1,K} \\ \vdots & \ddots & \vdots & \dots & \vdots & \ddots & \vdots \\ x_{1,1}y_{L,1} & \dots & x_{1,1}y_{L,K} & \dots & x_{1,M}y_{L,1} & \dots & x_{1,M}y_{L,K} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ x_{N,1}y_{1,1} & \dots & x_{N,1}y_{1,K} & \dots & x_{N,M}y_{1,1} & \dots & x_{N,M}y_{1,K} \\ \vdots & \ddots & \vdots & \dots & \vdots & \ddots & \vdots \\ x_{N,1}y_{L,1} & \dots & x_{N,1}y_{L,K} & \dots & x_{N,M}y_{L,1} & \dots & x_{N,M}y_{L,K} \end{bmatrix}$$

To implement Carleman approximation, the states of the system x are extended to

$$x_{\otimes} = [x^T x^{[2]T} \dots x^{[p]T}]^T, \quad (4)$$

where $x^{[p]} = x \otimes x^{[p-1]}$ denotes the p -th order Kronecker product of x .

For simplicity of presentation and without loss of generality we assume the nominal operating point is at the origin $x = 0$. Nonlinear vector functions $f(x)$, $g_j(x)$, $\gamma_l(x)$ are expanded in Maclaurin series of the following form:

$$f(x) = f(0) + \sum_{k=1}^{\infty} \frac{1}{k!} \frac{\partial f_{[k]}}{\partial x} \bigg|_{x=0} x^{[k]} \quad (5)$$

$$g_j(x) = g_j(0) + \sum_{k=1}^{\infty} \frac{1}{k!} \frac{\partial g_{j[k]}}{\partial x} \bigg|_{x=0} x^{[k]} \quad (6)$$

$$\gamma_l(x) = \gamma_l(0) + \sum_{k=1}^{\infty} \frac{1}{k!} \frac{\partial \gamma_{l[k]}}{\partial x} \bigg|_{x=0} x^{[k]} \quad (7)$$

Defining,

$$A_k = \frac{1}{k!} \frac{\partial f_{[k]}}{\partial x} \bigg|_{x=0}, \quad B_{jk} = \frac{1}{k!} \frac{\partial g_{j[k]}}{\partial x} \bigg|_{x=0}, \quad D_{lk} = \frac{1}{k!} \frac{\partial \gamma_{l[k]}}{\partial x} \bigg|_{x=0}, \quad (8)$$

$$A_0 = f(0), \quad B_{j0} = g_j(0), \quad D_{l0} = \gamma_l(0)$$

The nonlinear dynamic system (1)-(2) can be approximated by a polynomial form:

$$\dot{x} \cong \sum_{k=0}^p A_k x^{[k]} + \sum_{j=1}^m \sum_{k=0}^p B_{jk} x^{[k]} u_j + \sum_{l=1}^n \sum_{k=0}^p D_{lk} x^{[k]} w_l \quad (9)$$

$$h(x) = Cx \quad (10)$$

The polynomial order p is assumed to be high enough so that the truncation errors are negligible [19].

A notable detail: $\partial f_{[k]}$, $\partial g_{j[k]}$, $\partial \gamma_{l[k]}$ are derivatives based on the Kronecker product. For example, if $x = [x_1 \ x_2]$, $k = 2$,

$$\partial f_{[2]} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} \quad (11)$$

We also expand the coefficient matrices in (9) to obtain $\mathcal{A}$, $\mathcal{A}_0$, $\mathcal{B}_j$, $\mathcal{B}_{j0}$, $\mathcal{D}_l$, $\mathcal{D}_{l0}$ and $\mathcal{C}$, based on Kronecker products. They are composed in the following forms [17] [18]:

$$\mathcal{A} = \begin{bmatrix} A_{1,1} & A_{1,2} & \dots & A_{1,p} \\ A_{2,0} & A_{2,1} & \dots & A_{2,p-1} \\ 0 & A_{3,0} & \dots & A_{3,p-2} \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & A_{p,1} \end{bmatrix}, \quad \mathcal{A}_0 = \begin{bmatrix} A_{1,0} \\ 0 \\ 0 \\ \dots \\ 0 \end{bmatrix},$$

$$\mathcal{B}_j = \begin{bmatrix} B_{j1,1} & B_{j1,2} & \dots & B_{j1,p} \\ B_{j2,0} & B_{j2,1} & \dots & B_{j2,p-1} \\ 0 & B_{j3,0} & \dots & B_{j3,p-2} \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & B_{jp,1} \end{bmatrix}, \quad \mathcal{B}_{j0} = \begin{bmatrix} B_{j1,0} \\ 0 \\ 0 \\ \dots \\ 0 \end{bmatrix},$$

$$\mathcal{D}_l = \begin{bmatrix} D_{l1,1} & D_{l1,2} & \dots & D_{l1,p} \\ D_{l2,0} & D_{l2,1} & \dots & D_{l2,p-1} \\ 0 & D_{l3,0} & \dots & D_{l3,p-2} \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & D_{lp,1} \end{bmatrix}, \quad \mathcal{D}_{l0} = \begin{bmatrix} D_{l1,0} \\ 0 \\ 0 \\ \dots \\ 0 \end{bmatrix},$$

where $A_{k,i} = \sum_{m=0}^{k-1} I_n^{[m]} \otimes A_i \otimes I_n^{[k-1-m]}$, $B_{jk,i} = \sum_{m=0}^{k-1} I_n^{[m]} \otimes B_{ji} \otimes I_n^{[k-1-m]}$, $D_{lk,i} = \sum_{m=0}^{k-1} I_n^{[m]} \otimes D_{li} \otimes I_n^{[k-1-m]}$.

The following bilinear ODE has an expanded dimension and contains nonlinear dynamic information:

$$\dot{x}_{\otimes} = \mathcal{A}x_{\otimes} + \mathcal{A}_0 + \sum_{j=1}^m (\mathcal{B}_j x_{\otimes} + \mathcal{B}_{j0})u_j + \sum_{l=1}^n (\mathcal{D}_l x_{\otimes} + \mathcal{D}_{l0})w_l \quad (12)$$

$$y = \mathcal{C}x_{\otimes} + v \quad (13)$$

III. COMBINED ALGORITHM: CMHE/CMPC PAIR

To distinguish MHE and MPC, we use subscript E to denote estimation and subscript C to denote control.

A. CMHE design

MHE uses a sliding time window of past output trajectory, to update estimates in the past and to obtain the most probable current state. Since MHE uses discrete-time models, we assume that noise signals can be discretized in time and affect the system as piecewise constant noise signals.

$$\begin{aligned} \hat{x}_0^*, \hat{w}^* = \arg \min_{\hat{x}_0, \hat{w}} J_E &= \Lambda(\hat{x}_0) + \sum_{i=1}^{N_E} L_i(\hat{w}_i, \hat{v}_i) \quad (14) \\ \forall i &= 1, \dots, N_E \\ \text{s.t.} \end{aligned}$$

$$\hat{x}_i \in \mathcal{X}, \hat{w}_i \in \mathcal{D}, \hat{v}_i \in \mathcal{V} \quad (15)$$

$$\hat{v}_i = y_i - \hat{y}_i \quad (16)$$

$$\Lambda(\hat{x}_0) = (\bar{x}_0 - \hat{x}_0)^T \Pi_0 (\bar{x}_0 - \hat{x}_0) \quad (17)$$

$$L_i(\hat{w}_i, \hat{v}_i) = \hat{v}_i^T Q_E \hat{v}_i + \hat{w}_i^T R_E \hat{w}_i \quad (18)$$

In this dynamic optimization problem, the design variables are $\hat{x}_0$, the state estimate at the beginning of the sliding window, and $\hat{w}$, the matrix of noise estimates, with $\hat{w}_i$ the estimate of weight averaged noise vector in the i -th time interval. $\mathcal{X}$, $\mathcal{D}$, and $\mathcal{V}$ are compact sets. $\hat{v}_i$ is the estimation error. $\Lambda(\hat{x}_0)$ is the arrival cost, which summarizes the past data before the sliding time window. We pick the estimate of state $\hat{x}_0$ at the previous sampling time to represent $\bar{x}_0$, which is one step before the sliding window. $L_i, \forall i = 1, \dots, N_E$ denotes the stage costs. Q_E and R_E are weighting matrices that have symmetric positive definite structures.

With control signals u continuously generated by CMPC (details presented in section III-B), the u used in the CMHE window is updated as the window recedes, increasing the precision of the estimation. Fig. 1 presents a schematic of CMHE/CMPC pair. At T_{k-1} , CMHE uses a window of $[T_{k-4}, T_{k-1}]$ (dashed red) to estimate $\hat{x}_{\otimes, k-1}$ and $[\hat{w}_{k-3}, \hat{w}_{k-2}, \hat{w}_{k-1}]$. In this window, $[u_{k-3}, u_{k-2}, u_{k-1}]$, provided by the CMPC from previous steps, are used as the inputs in the dynamic model. With $\hat{x}_{\otimes, k-1}$ as the initial condition, CMPC uses a window of $[T_{k-1}, T_{k+3}]$ (dashed blue) to compute the input sequence $[u_k, u_{k+1}, u_{k+2}, u_{k+3}]$. Discarding $u_{k+1}, u_{k+2}, u_{k+3}$, only u_k is implemented to the system. As time proceeds on to T_k , u_k is supplied to CMHE with a window of $[T_{k-3}, T_k]$ (black), updating the inputs to $[u_{k-2}, u_{k-1}, u_k]$.

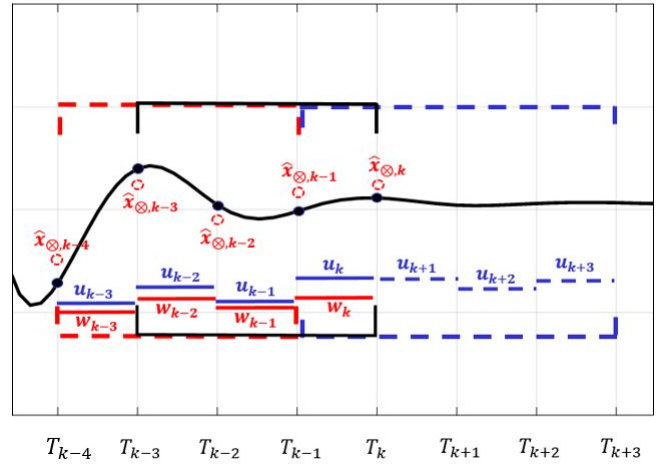


Fig. 1. A schematic of CMHE/CMPC pair time windows and variables.

Consequently, the following model is used in solving the MHE problem of (14)-(18)

$$\begin{aligned} \dot{\hat{x}}_{\otimes} &= \mathcal{A}\hat{x}_{\otimes} + \mathcal{A}_0 + \sum_{j=1}^m (\mathcal{B}_j \hat{x}_{\otimes} + \mathcal{B}_{j0})u_{j,i} \\ &+ \sum_{l=1}^n (\mathcal{D}_l \hat{x}_{\otimes} + \mathcal{D}_{l0})\hat{w}_{l,i} \quad (19) \end{aligned}$$

$$\hat{y}_i = \mathcal{C}\hat{x}_{\otimes, i} + \hat{v}_i \quad (20)$$

$$\hat{x}_{\otimes, 0} = \hat{x}_{\otimes}(t_0) \quad (21)$$

Note, most computations of (19)-(21) have already been performed by the CMPC at the previous steps and are provided to the MHE, significantly accelerating this part of the computations!

B. CMPC design

The optimal control problem is recast as a recursion of receding finite-horizon dynamic optimization problems at every time point t_0 , which has a general form:

$$u^* = \arg \min_u \int_{t_0}^{t_f} J_C(x, u, w) dt \quad (22)$$

$$u_j(t) = \sum_{i=1}^{N_C} u_{j,i} \mathbf{B}(t; T_{i-1}, T_i), \forall j = 1, \dots, m \quad (23)$$

$$T_0 = t_0, T_{N_C} \leq t_f \quad (24)$$

$$\dot{x} - f(x) - \sum_{j=1}^m g_j(x)u_j(t) - \sum_{l=1}^n \gamma_l(x)w_l(t) = 0 \quad (25)$$

$$x(t_0) = x_0 \quad (26)$$

$$f^c(x, u) \leq 0 \quad (27)$$

where J_C is the cost function of CMPC and f^c denotes a vector function of equality and inequality constraints. Similarly to CMHE, each manipulated input is discretized in time; u denotes the matrix of manipulated inputs with u_j being the j -th manipulated input vector consisting of $u_{j,i}, \forall i = 1, \dots, N_C$. $u_{j,i}$ is the i -th piecewise constant value of u_j that takes place in the i -th time interval $(T_{i-1}, T_i]$. The i -th time interval is also defined as the i -th sampling time and has a length of ΔT_i . We

define $\mathbf{B}(t; T_{i-1}; T_i) = \mathbf{H}(t - T_{i-1}) - \mathbf{H}(t - T_i)$ as a rectangular pulse function, where $\mathbf{H}$ is the standard Heaviside function. T_{i-1} and T_i denote the initiation time and the termination time respectively.

T_0 is the beginning and T_{N_C} is the end of the control horizon respectively. $t_0 \equiv T_0$ is the beginning and t_f is the end of the prediction horizon respectively. The length of the prediction horizon can be equal to or greater than the control horizon, depending on the requirement for controller robustness and stability [25][26].

When MPC predicts the system evolution, the unknown noise in the future is not yet realized. Thus, w or $w_l, l = 1, \dots, n$ are assumed to be zero. The cost function $J_C(x, u, w)$ is reduced to $J_C(x, u)$, which can be approximated through differentiation by the following:

$$J_C(x, u) = \sum_{k=0}^{\infty} \sum_{l=0}^{\infty} \frac{1}{(k+l)!} \frac{\partial^{[k+l]} J_C}{\partial x^k \partial u^l} \Big|_0 x^k u^l \quad (28)$$

For some cases, the time integral $\int_{T_0}^{T_N} J_C(x, u) dt$ is a better choice as the cost function, especially in regulation or tracking problems.

$$\begin{aligned} \int_{T_0}^{T_N} J_C(x, u) dt &\cong J_0(T_N - T_0) + \sum_{i=1}^N \sum_{j=1}^m J_{Bj} u_{\otimes, j, i} \Delta T_i \\ &+ \sum_{i=1}^N (J_A + \sum_{j=1}^m J_{Nj} u_{\otimes, j, i} \otimes) \int_{T_{i-1}}^{T_i} x_{\otimes} dt \end{aligned} \quad (29)$$

where J_0, J_A, J_{Bj} and J_{Nj} are Jacobian matrices.

C. Analytical Prediction of System Evolution

We define the following notations for the purpose of simplicity in the derivations:

$$\tilde{\mathcal{A}}_i = \mathcal{A} + \sum_{j=1}^m \mathcal{B}_j u_{j,i} + \sum_{l=1}^n \mathcal{D}_l w_{l,i}, \quad (30)$$

$$\mathcal{E}_i = \exp(\tilde{\mathcal{A}}_i \Delta T_i), \quad (31)$$

$$\mathcal{G} u U_i = \tilde{\mathcal{A}}_i^{-1} [\mathcal{E}_i - I], \quad (32)$$

$$\mathcal{F}_i = \mathcal{A}_0 + \sum_{j=1}^m \mathcal{B}_{j0} u_{j,i} + \sum_{l=1}^n \mathcal{D}_{l0} w_{l,i} \quad (33)$$

As presented earlier, the nonlinear system is expressed in a bilinear form of(12). We use the convolution integral to obtain an analytical solution in the form of

$$\begin{aligned} x(t)_{\otimes} &= \exp \left[\mathcal{A}_i(t - T_{i-1}) \right] x(T_{i-1})_{\otimes} \\ &+ \int_{T_{i-1}}^t \exp \left[\mathcal{A}_i(t - \tau) \right] d\tau \cdot \mathcal{F}_i \end{aligned} \quad (34)$$

This analytical solution is used in the CMHE/CMPC pair, reducing the dynamic constraints in the optimization. The cost functions J_E and $\int_{T_0}^{T_N} J_C(x, u) dt$ are also calculated analytically. Furthermore, they enable us to provide analytical gradient vector and Hessian matrix expressions to facilitate the search process. These analytical calculations significantly accelerate the optimal search. Their derivation is presented in the next section.

D. Gradient vector and Hessian matrix derivations

In this section, detailed expressions to calculate the Gradient vector and the Hessian matrix of the CMHE/CMPC pair are presented.

1) *CMHE part:* At the k -th sampling time, the gradient vector of CMHE has the following form:

$$G_E = \left[\frac{\partial J_E}{\partial \hat{x}_{0,k}} \quad \frac{\partial J_E}{\partial \hat{w}_{1,k}} \quad \frac{\partial J_E}{\partial \hat{w}_{2,k}} \quad \dots \quad \frac{\partial J_E}{\partial \hat{w}_{p,k}} \right]; \quad (35)$$

The elements to compose the gradient vector are:

$$\frac{\partial J_E}{\partial \hat{x}_{0,k}} = 2(x_0 - \hat{x}_0)^T \Pi_0 \Gamma_k + \sum_{i=1}^N 2v_i^T Q_E \left(\frac{\partial v_i}{\partial \hat{x}_{0,k}} \right) \quad (36)$$

and

$$\frac{\partial J_E}{\partial \hat{w}_{p,k}} = 2\hat{w}_k^T R_E \Gamma_k + \sum_{i=j}^N 2v_i^T Q_E \left(\frac{\partial v_i}{\partial \hat{w}_{p,k}} \right) \quad (37)$$

where Γ_k is a vector with zero elements except the k -th element which has the unity value.

By differentiating Eq. (35), the Hessian matrix obtains the following symmetric form:

$$H_E = \begin{bmatrix} \frac{\partial^2 J_E}{\partial \hat{x}_{0,k} \partial \hat{x}_{0,k}} & \frac{\partial^2 J_E}{\partial \hat{x}_{0,k} \partial \hat{w}_{1,k}} & \dots & \frac{\partial^2 J_E}{\partial \hat{x}_{0,k} \partial \hat{w}_{p,k}} \\ \dots & \frac{\partial^2 J_E}{\partial \hat{w}_{1,k} \partial \hat{w}_{1,k}} & \dots & \frac{\partial^2 J_E}{\partial \hat{w}_{1,k} \partial \hat{w}_{p,k}} \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \frac{\partial^2 J_E}{\partial \hat{w}_{p,k} \partial \hat{w}_{p,k}} \end{bmatrix}$$

By differentiating Eq. (36) and (37), the Hessian matrix elements are given by the following equations:

$$\begin{aligned} \frac{\partial^2 J_E}{\partial \hat{x}_{0,l} \partial \hat{x}_{0,k}} &= 2\Gamma_l^T \Pi_0 \Gamma_k + 2 \sum_{i=1}^N \left(\frac{\partial v_i}{\partial \hat{x}_{0,k}} \right)^T Q_E \left(\frac{\partial v_i}{\partial \hat{x}_{0,l}} \right) \\ &+ 2 \sum_{i=1}^N v_i^T Q_E \left(\frac{\partial^2 v_i}{\partial \hat{x}_{0,l} \partial \hat{x}_{0,k}} \right) \end{aligned} \quad (38)$$

$$\begin{aligned} \frac{\partial^2 J_E}{\partial \hat{x}_{0,l} \partial \hat{w}_{p,k}} &= 2 \sum_{i=1}^N \left(\frac{\partial v_i}{\partial \hat{x}_{0,l}} \right)^T Q_E \left(\frac{\partial v_i}{\partial \hat{w}_{p,k}} \right) \\ &+ 2 \sum_{i=1}^N v_i^T Q_E \left(\frac{\partial^2 v_i}{\partial \hat{x}_{0,l} \partial \hat{w}_{p,k}} \right) \end{aligned} \quad (39)$$

$$\begin{aligned} \frac{\partial^2 J_E}{\partial \hat{w}_{q,l} \partial \hat{w}_{p,k}} &= 2\Gamma_l^T R_E \Gamma_k + 2 \sum_{i=1}^N \left(\frac{\partial v_i}{\partial \hat{w}_{q,l}} \right)^T Q_E \left(\frac{\partial v_i}{\partial \hat{w}_{p,k}} \right) \\ &+ 2 \sum_{i=1}^N v_i^T Q_E \left(\frac{\partial^2 v_i}{\partial \hat{w}_{q,l} \partial \hat{w}_{p,k}} \right) \end{aligned} \quad (40)$$

From Clairaut's theorem, we can conclude the second derivatives are symmetric. Therefore, from (36)-(40) all elements of the Hessian matrix are now available.

2) *CMPC part*: The gradient vector of CMPC, i.e. the sensitivity of $\int_{T_0}^{T_N} J_C dt$ to the manipulated input vector, for the k -th manipulated input vector $u_k = [u_{k,1}, u_{k,2}, \dots, u_{k,p}]$ is

$$G_k = \left[\frac{\partial}{\partial u_{k,1}} \int_{T_0}^{T_N} J_C dt \quad \dots \quad \frac{\partial}{\partial u_{k,p}} \int_{T_0}^{T_N} J_C dt \right] \quad (41)$$

The $\mathcal{K}$ -th element in the gradient vector above is:

$$\begin{aligned} \frac{\partial}{\partial u_{k,\mathcal{K}}} \int_{T_0}^{T_N} J_C dt &= (J_A + \sum_{j=1}^m J_{Nj} u_{\otimes,j,\mathcal{K}} \otimes) \sum_{i=\mathcal{K}}^N \int_{T_{i-1}}^{T_i} \frac{\partial x_{\otimes}}{\partial u_{k,\mathcal{K}}} dt \\ &+ J_{Nk} (\partial u_{\otimes,k,\mathcal{K}}) \otimes \int_{T_{\mathcal{K}-1}}^{T_{\mathcal{K}}} x_{\otimes} dt + J_{Bk} (\partial u_{\otimes,k,\mathcal{K}}) \Delta T_{\mathcal{K}} \end{aligned} \quad (42)$$

By differentiating the gradient vector of (41), the Hessian matrix attains the following symmetric form:

$$H_k = \begin{bmatrix} \frac{\partial^2}{\partial^2 u_{k,1}} \int_{T_0}^{T_N} J_C dt & \dots & \frac{\partial^2}{\partial u_{k,1} \partial u_{k,p}} \int_{T_0}^{T_N} J_C dt \\ \dots & \dots & \frac{\partial^2}{\partial u_{k,2} \partial u_{k,p}} \int_{T_0}^{T_N} J_C dt \\ \dots & \dots & \dots \\ \dots & \dots & \frac{\partial^2}{\partial^2 u_{k,p}} \int_{T_0}^{T_N} J_C dt \end{bmatrix} \quad (43)$$

The Hessian matrix elements are given by the following equations:

$$\begin{aligned} \frac{\partial^2}{\partial^2 u_{k,\mathcal{K}}} \int_{T_0}^{T_N} J_C dt &= (J_A + \sum_{j=1}^m J_{Nj} u_{\otimes,j,\mathcal{K}} \otimes) \sum_{i=\mathcal{K}}^N \int_{T_{i-1}}^{T_i} \frac{\partial^2 x_{\otimes}}{\partial^2 u_{k,\mathcal{K}}} dt \\ &+ 2J_{Nk} \partial u_{\otimes,k,\mathcal{K}} \otimes \sum_{i=\mathcal{K}}^N \int_{T_{i-1}}^{T_i} \frac{\partial x_{\otimes}}{\partial u_{k,\mathcal{K}}} dt \\ &+ J_{Nk} (\partial^2 u_{\otimes,k,\mathcal{K}}) \otimes \int_{T_{\mathcal{K}-1}}^{T_{\mathcal{K}}} x_{\otimes} dt + J_{Bk} (\partial^2 u_{\otimes,k,\mathcal{K}}) \Delta T_{\mathcal{K}} \end{aligned} \quad (44)$$

For $\mathcal{K} < q < N$

$$\begin{aligned} \frac{\partial^2}{\partial u_{k,q} \partial u_{k,\mathcal{K}}} \int_{T_0}^{T_N} J_C dt &= \\ &(J_A + \sum_{j=1}^m J_{Nj} u_{\otimes,j,\mathcal{K}} \otimes) \sum_{i=\mathcal{K}}^N \int_{T_{i-1}}^{T_i} \frac{\partial^2 x_{\otimes}}{\partial u_{k,q} \partial u_{k,\mathcal{K}}} dt \\ &+ J_{Nk} \partial u_{\otimes,k,q} \otimes \sum_{i=\mathcal{K}}^N \int_{T_{i-1}}^{T_i} \frac{\partial x_{\otimes}}{\partial u_{k,\mathcal{K}}} dt \end{aligned} \quad (45)$$

IV. ILLUSTRATIVE EXAMPLE

We use a nonlinear jacketed continuously-stirred tank reactor (CSTR) as a simulation example [27] to demonstrate the effectiveness of the proposed CMHE/CMPC pair. Reactant A is fed to the reactor where an exothermic first-order reaction takes place. The temperature inside the reactor is manipulated by using a jacket to the reactor with coolant water. The dynamics of this system are mathematically obtained by applying a material balance for the reactant and an energy balance and are described with the following two ODEs:

$$\begin{aligned} \dot{C}_A &= \frac{q+w}{V} (C_{Af} - C_A) - k_0 \exp\left(-\frac{E}{R T_R}\right) C_A \\ \dot{T}_R &= \frac{q+w}{V} (T_f - T_R) - \frac{\Delta H}{\rho C_p} k_0 \exp\left(-\frac{E}{R T_R}\right) C_A \\ &\quad + \frac{UA}{V \rho C_p} (T_c - T_R) \end{aligned} \quad (46)$$

TABLE I

UNSTABLE CSTR PARAMETERS AND NOMINAL STEADY STATE

Parameter	Description	Value
q	Feed Flow Rate	100 L/min
C_{Af}	Feeder Concentration	1 gmol/L
T_f	Feed Temperature	350 K
V	Reactor Volume	100 L
UA	Heat Transfer Coefficient	5×10^4 J/(min·K)
k_0	Exponential Factor	7.2×10^{10} min $^{-1}$
E/R	Reduced Activation Energy	8750 K
ΔH	Heat of Reaction	-5×10^4 J/mol
ρ	Density of Reactor Contents	1000 g/L
C_p	Heat Capacity of Reactor Contents	0.239 J/(g·K)
T_c	Coolant Temperature	311.1 K
C_A	Nominal Concentration of Reactor Contents	9.3413×10^{-2} gmol/L
T_R	Reactor Temperature	385 K

The manipulated input is the coolant water temperature T_c . The feeding flow rate q is affected by unknown noise w . The reactor state variables are the reactant concentration, C_A , and temperature, T_R . In this illustration, T_R is measurable while C_A is unavailable. CMHE first estimates C_A by measuring T_R , and CMPC then computes the control signals based on the state estimate to form a closed-loop system.

Table I lists the parameters and the nominal operating condition. The system is highly nonlinear around the nominal steady state listed in Table I and a cursory analysis deduces the CSTR is unstable at the current operating point. We apply 2nd order Carleman approximation around the nominal steady state listed in Table I, to model the system and design the CMHE/CMPC pair. Carleman matrices are computed off-line, and need to be computed only one time.

In this simulation example, we assume the unmeasured noise w affecting the flow rate q , is a uniformly distributed white noise, which is bounded in $[-10, 10]$ L/min. In addition, there exists sensor noise affecting the measurement of T_R . It also has a uniform distribution and is bounded in $[-0.02, 0.02]$ K. The sampling time for both CMHE and CMPC is 0.05 min. The sliding estimation window has a horizon of $N_E = 3$. The CMHE design uses a quadratic cost function

$$\{\hat{x}_0^*, \hat{w}_1^*, \dots, \hat{w}_{N_E}^*\} = \arg \min_{\hat{x}_0, \hat{w}, \hat{v}} J_E$$

$$J_E = (\bar{x}_0 - \hat{x}_0)^T \Pi_0 (\bar{x}_0 - \hat{x}_0) + \sum_{i=1}^{N_E} (\hat{v}_i^T Q_E \hat{v}_i + \hat{w}_i^T R_E \hat{w}_i) \quad (47)$$

where $\hat{x}_0$, $\hat{w}_i$, $\hat{v} \forall i = 1, 2, 3$ are the design variables, subject to the dynamics constraint (46) and box constraints:

$$\begin{aligned} 0.08 \leq C_A \leq 0.11, \quad -10 \leq \hat{w}_i \leq 10, \quad -0.02 \leq \hat{v}_i \leq 0.02 \\ \forall i = 1, 2, 3 \end{aligned} \quad (48)$$

and weights

$$Q_E = 1, R_E = \begin{bmatrix} 10^{-5} & 0 \\ 0 & 10^{-5} \end{bmatrix}, \Pi_0 = \begin{bmatrix} 10^{-6} & 0 \\ 0 & 10^{-6} \end{bmatrix}.$$

The control objective is to regulate the system at the reactor temperature of $T_R = 385$ K, under process noise w affecting q . On one side, CMHE continuously estimates the current state at each decision time for CMPC to make control decisions for the next time period. On the other side, CMPC continuously updates the dynamic model in the last period used by CMHE to increase accuracy.

The CMPC design also uses the following quadratic cost functional:

$$u^* = \arg \min_u \int_{T_0}^{T_{N_p}} J_C(x, u) dt = \sum_{i=1}^{N_p} \int_{T_{i-1}}^{T_i} (x^T Q_C x + u^T R_C u) dt$$

$$\cong J_0(T_{N_p} - T_0) + \sum_{i=1}^{N_p} J_A \int_{T_{i-1}}^{T_i} x_{\otimes} dt + \sum_{i=1}^{N_p} \sum_{j=1}^m J_{Bj} u_{\otimes, j, i} \Delta T_i$$

where u is the control variable, subject to the dynamic constraints of (46). The prediction horizon is chosen as $N_p = 8$ and the control horizon as $N_C = 4$. The weights are chosen to be $Q_C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $R_C = 10^{-5}$. The cost function is reconstructed with extended states $x_{\otimes}$ and extended control signals $u_{\otimes}$. Correspondingly, the coefficient vectors J_0 , J_A and J_B have dimensions that are consistent with $x_{\otimes}$ and $u_{\otimes}$.

All simulations were performed using a windows PC with 3.4 GHz Intel Core i7 processor. With standard NMHE/NMPC pair, the average computation time for one control signal is 3.40 s, which exceeds the length of one sampling time 0.05 min (≈ 3 s) by 113%. As mentioned in the introduction, it may cause significant performance degradation and safety issues. With the CMHE/CMPC pair, the analytical solutions reduce the average computation time to 0.65 s or 22% of one sampling time. With the Gradient vector and the Hessian matrix supplied analytically, the CMHE/CMPC pair further reduced the computation time to 0.22 s or 7.3% of one sampling time.

Fig. 2 and Fig. 3 present the closed-loop performance under NMHE/NMPC pair (green lines) and CMHE/CMPC pair (blue lines), respectively. They are both compared with the open-loop response under the same process noise (red lines). The open-loop system goes through a periodic orbit under the effect of noise. We observe that both NMHE/NMPC and CMHE/CMPC pairs are able to suppress the effect of noise and stabilize the closed-loop system at the desired set-point. It is important to observe that in Fig. 2 and Fig. 3, the closed-loop performance under NMHE/NMPC pair (green lines) and CMHE/CMPC pair (blue lines) are almost identical; the proposed CMHE/CMPC pair though achieves the same performance at only a fraction of the computational cost (from 3.40 s to 0.22 s or 15.5 times faster), which illustrates the computational efficiency of the proposed pair.

V. CONCLUSIONS

An optimal output feedback controller was proposed that combines optimal estimators and optimal controllers. The proposed structure combines a moving horizon estimator and model predictive controller to circumvent complex formulas and implicitly respect constraints. It attempts to address the

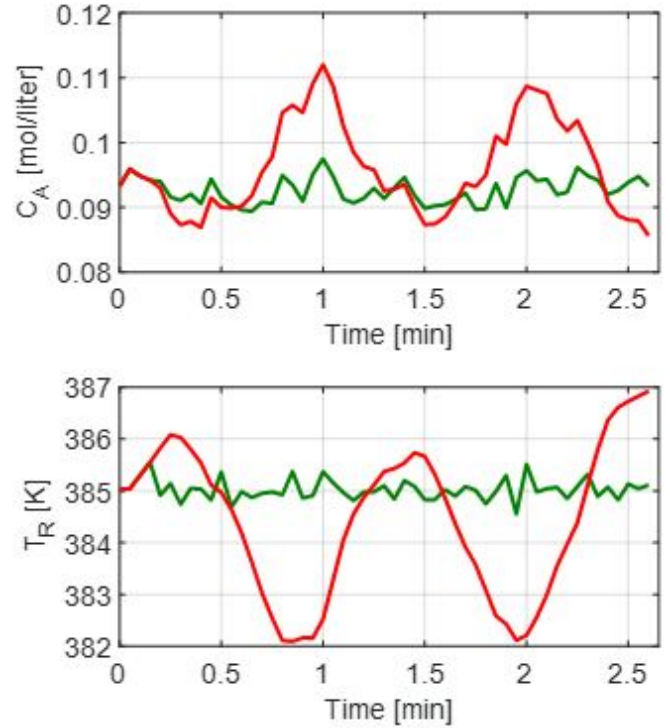


Fig. 2. Closed loop response under NMHE/NMPC pair (green lines) compared with open loop response of the system (red lines): concentration of reactor contents C_A , reactor temperature T_R

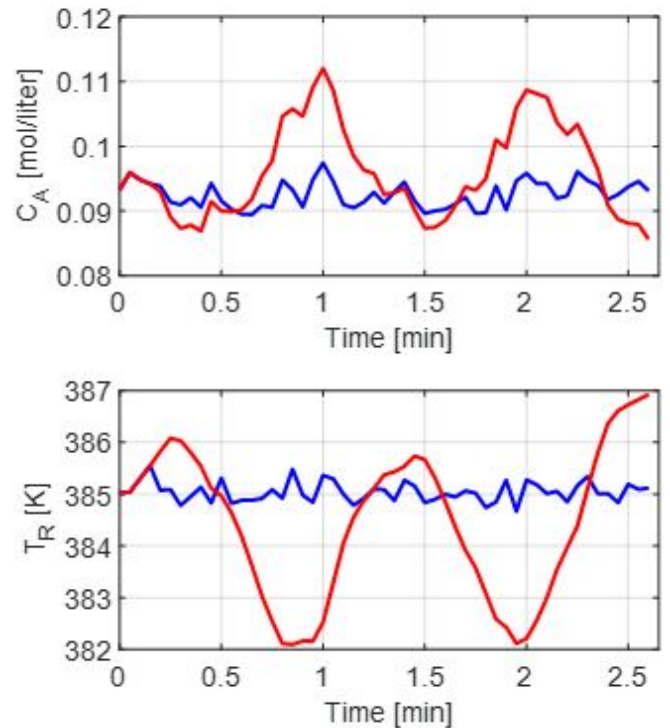


Fig. 3. Closed loop response under 2nd order CMHE/CMPC pair (blue lines) compared with open loop response of the system (red lines): concentration of reactor contents C_A , reactor temperature T_R

overarching computational hurdle of the associated optimization algorithms by employing Carleman approximation. It uses an extended bilinear process model to predict system evolution analytically. It replaces the dynamic constraints in the optimization with analytically computed nonlinear ones. Analytical Gradient vectors and Hessian matrices are supplied to the underlying the search algorithms to the CMHE/CMPC pair to accelerate the convergence to the optimal solution. This is illustrated in a simulation example where the computation time was reduced by a factor of 15 times, from 113 % of one sampling time to 22% and 7.3%. In future work, advanced-step NMHE/NMPC theories will be applied to the CMHE/MPC pair, further accelerating computations. The analytical sensitivities of the Carleman expression will mesh particularly well with advanced step ones. The stability properties of the CMHE/CMPC pair will also be investigated.

APPENDIX

The following derivations are the components of the above equations:

$$u_{\otimes,k,\mathcal{K}} = [u_{k,\mathcal{K}}, u_{k,\mathcal{K}}^2, \dots, u_{k,\mathcal{K}}^p] \quad (49)$$

$$\partial u_{\otimes,k,\mathcal{K}} = [1, 2 u_{k,\mathcal{K}}, \dots, p u_{k,\mathcal{K}}^{p-1}] \quad (50)$$

$$\partial^2 u_{\otimes,k,\mathcal{K}} = [0, 2, \dots, p(p-1) u_{k,\mathcal{K}}^{p-2}] \quad (51)$$

$$\frac{\partial x_{\otimes,i}}{\partial x_{\otimes,i-1}} = \mathcal{E}_i \quad (52)$$

$$\int_{T_{i-1}}^{T_i} \frac{\partial x_{\otimes}}{\partial x_{\otimes,i-1}} dt = \mathcal{G} u U_i \quad (53)$$

$$\begin{aligned} \frac{\partial x_{\otimes,\mathcal{K}}}{\partial u_{k,\mathcal{K}}} &= \frac{\partial \mathcal{E}_{\mathcal{K}}}{\partial u_{k,\mathcal{K}}} x_{\otimes,\mathcal{K}-1} \\ &+ \tilde{\mathcal{A}}_{\mathcal{K}}^{-1} \frac{\partial \mathcal{E}_{\mathcal{K}}}{\partial u_{k,\mathcal{K}}} \mathcal{F}_{\mathcal{K}} + \mathcal{G} u(U_{\mathcal{K}}) \mathcal{B}_{k0} \\ &- \tilde{\mathcal{A}}_{\mathcal{K}}^{-1} \mathcal{B}_k \mathcal{G} u(U_{\mathcal{K}}) \mathcal{F}_{\mathcal{K}} \end{aligned} \quad (54)$$

$$\begin{aligned} \int_{T_{\mathcal{K}-1}}^{T_{\mathcal{K}}} \frac{\partial x_{\otimes}}{\partial u_{k,\mathcal{K}}} dt &= \int_{T_{\mathcal{K}-1}}^{T_{\mathcal{K}}} \frac{\partial \mathcal{E}_{\mathcal{K}}}{\partial u_{k,\mathcal{K}}} dt \cdot x_{\otimes,\mathcal{K}-1} \\ &+ \tilde{\mathcal{A}}_{\mathcal{K}}^{-1} \int_{T_{\mathcal{K}-1}}^{T_{\mathcal{K}}} \frac{\partial \mathcal{E}_{\mathcal{K}}}{\partial u_{k,\mathcal{K}}} dt \cdot \mathcal{F}_{\mathcal{K}} \\ &+ \tilde{\mathcal{A}}_{\mathcal{K}}^{-1} [\mathcal{G} u(U_{\mathcal{K}}) - \Delta T_{\mathcal{K}} \cdot I] \mathcal{B}_{k0} \\ &- \tilde{\mathcal{A}}_{\mathcal{K}}^{-1} \mathcal{B}_k \tilde{\mathcal{A}}_{\mathcal{K}}^{-1} [\mathcal{G} u(U_{\mathcal{K}}) - \Delta T_{\mathcal{K}} \cdot I] \mathcal{F}_{\mathcal{K}} \end{aligned} \quad (55)$$

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